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Multi-resolution Copernicus DSM-based cumulative viewshed assessment of antenna towers for wildfire monitoring in Montenegro

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Abstract: Wildfires are an increasingly significant environmental and socio-economic problem in Mediterranean and mountainous regions. This study evaluates the wildfire surveillance potential of an existing commercial antenna tower network in Montenegro, an infrastructure designed for telecommunications rather than terrain visibility, using a multi-scenario cumulative viewshed approach. Ninety-nine antenna towers operated by One Crna Gora were analysed as potential locations for wildfire monitoring cameras. The methodology integrated Geographic Information Systems (GIS), automated processing with the Geospatial Data Abstraction Library (GDAL), and Copernicus Digital Surface Models (DSM) at 10, 30 and 90 m resolution. In total, 108 cumulative scenarios were generated by varying observer height (10, 20, 30 m), target height (0, 20, 50, 100 m) representing successive wildfire development stages, and visibility radius (5, 10, 15 km). Coverageranged from 8.6 to 58.5 % of the national territory. Target height proved the dominant control, while the effect of DSM resolution was strongly conditional: the 90 m dataset overestimated coverage by up to 41 % relative to the 10 m dataset at ground level, but by only about 3 % for a 100 m smoke plume. This dependence, quantified here across the full parameter space, indicates that resolution requirements should follow from the intended detection objective. The results provide a preliminary geometric basis for wildfire monitoring planning in complex mountainous Mediterranean environments.

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1 Introduction

Recent analyses indicate a continuous increase in the frequency and intensity of wildfires worldwide, with climate change and anthropogenic pressures identified as the primary drivers of this trend [1–4]. Under such conditions, preventive wildfire risk management has become a strategic priority, requiring systematic planning, spatially targeted mitigation measures, and the strengthening of ecosystem and societal resilience [5]. Although different countries apply diverse methodological approaches and datasets in wildfire assessments [6], the growing complexity of wildfire regimes highlights the need for integrated and advanced prevention and monitoring models.

Traditional wildfire surveillance systems were primarily based on expert knowledge, field reconnaissance, and subjective assessments of terrain visibility when selecting observation tower locations. Although widely applied, such approaches were time-consuming and lacked systematic optimization methods capable of ensuring efficient spatial coverage. Since the effectiveness of wildfire detection systems directly depends on the placement of observation towers and camera-based monitoring infrastructure, inadequate positioning can significantly reduce early warning capabilities and delay response actions during wildfire events [7].

In recent years, the development of Geographic Information Systems (GIS) and of digital elevation datasets has enabled more innovative, spatially explicit, and data-driven approaches [8–10] for wildfire surveillance planning [7]. Two types of such datasets are commonly distinguished. A Digital Elevation Model (DEM), in the narrow sense of a bare-earth terrain model, represents the ground surface with vegetation and structures removed, whereas a Digital Surface Model (DSM) represents the reflective surface actually sensed, including forest canopy, buildings and other above-ground features. The distinction is decisive for

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visibility modelling, since a line of sight obstructed by a forest canopy is registered on a DSM but not on a bare-earth DEM, and canopy is the dominant above-ground barrier in forested Mediterranean landscapes. One of the most widely applied GIS techniques in this context is viewshed analysis, which evaluates line-of-sight visibility from one or more observation points while accounting for terrain barriers such as ridges, valleys, and complex mountainous topography [11–13]. Cumulative viewshed analysis extends this principle to networks of observation points and thereby enables the assessment of existing infrastructure, such as fire lookout towers, antenna towers, and telecommunication facilities, for wildfire monitoring purposes [14, 15]. By evaluating visibility coverage and identifying terrain shadow zones, this approach supports the optimization of surveillance networks and helps determine locations where additional observation points could improve early wildfire detection efficiency, while the use of existing infrastructure provides a cost-effective solution that reduces the need for constructing new facilities [16–19].

The process of conducting viewshed analysis consists of several stages, each of which introduces the possibility of generating errors that may affect the final visibility model [16]. These uncertainties arise primarily from the limitations of the applied technology and the inherent difficulty of accurately representing the Earth's surface in a digital environment. The factors influencing viewshed analysis accuracy can be classified into several categories: (a) DEM and DSM related errors and uncertainties [20], (b) the selection of spatial resolution [21, 22], (c) the choice of visibility algorithms and computational approaches [23], (d) parameter settings within the GIS environment [24], (e) external environmental factors such as vegetation, atmospheric conditions, and infrastructure [16], and (f) the sensitivity and detection parameters of wildfire surveillance cameras [21]. Together, these factors can significantly influence line-of-sight calculations, visibility estimation, and the overall reliability of wildfire detection and monitoring systems.

In addition to visibility coverage, wildfire monitoring systems must also consider operational and economic constraints associated with the construction and maintenance of surveillance infrastructure [17, 25]. Consequently, the use of existing infrastructure such as telecommunication facilities, antenna towers, and fire lookout towers has emerged as an important strategy for reducing implementation costs while maximizing monitoring efficiency [14, 15]. Recent studies have shown that the integration of GIS-based optimization methods, viewshed modelling, and existing infrastructure networks can significantly improve wildfire surveillance planning [20, 26] and support the development of more resilient wildfire early warning systems in fire-prone regions [25, 27].

Previous wildfire visibility studies have typically relied on a single DEM or DSM resolution and a fixed set of observation parameters, so that the effect of terrain representation and of wildfire development stage on detection potential has rarely been assessed jointly. In real wildfire conditions, however, visibility is highly dynamic and strongly influenced by the wildfire development stage, particularly the height of the smoke plume. Moreover, the surveillance networks examined have almost exclusively comprised purpose-built fire lookout towers, while the potential of existing commercial telecommunication infrastructure has remained largely unexplored, despite offering nationwide coverage at negligible construction cost. Therefore, this study aims to evaluate the wildfire surveillance potential of the existing antenna tower network in Montenegro using a multi-scenario cumulative viewshed approach. Montenegro was selected as the study area due to its high wildfire susceptibility and the large proportion of burned areas recorded in recent years [28, 29]. According to data from the European Forest Fire Information System (EFFIS) [30, 31], Montenegro has one of the highest burned-area proportions relative to its national territory in Europe, emphasizing the importance of efficient wildfire monitoring systems within this mountainous Mediterranean environment.

In order to provide a more realistic and operationally relevant assessment, three DSM resolutions (10, 30 and 90 m) were used together with multiple visibility scenarios generated by varying observer height (offset A = 10, 20 and 30 m), target height (offset B = 0, 20, 50 and 100 m), and visibility radius (5, 10 and 15 km). The selected offset B scenarios simulate different wildfire development stages, ranging from small ground-level fires to large and intense wildfires characterized by elevated smoke plumes visible over greater distances. This design directly addresses two of the limitations outlined above: the sensitivity of visibility estimation to DSM resolution is quantified rather than assumed, and the use of a surface model together with variable target heights reduces the oversimplification inherent in binary, ground-level visibility assessments.

2 Study area

The analysis covers the entire territory of Montenegro (Figure 1), a country of 13,883 km^2 on the eastern Adriatic coast of the Balkan Peninsula. Mountains exceeding 2,500 m

rise within less than 40 km of the sea: only 10.3 % of the territory lies below 200 m, while 56.9 % exceeds 1,000 m. Vertical differences of more than 1,000 m occur repeatedly over short horizontal distances, so that ridges, incised canyons and enclosed depressions generate extensive terrain shadow zones irrespective of the elevation of the observation point. Carbonate rocks underlie more than 60 % of the surface, and the resulting karst landforms operate at scales of tens to a few hundred metres, precisely the range at which surface model resolution determines whether an obstruction is represented or smoothed away. Forests and forest land occupy 59.6 % of the national territory, with the coastal belt dominated by pine stands and evergreen maquis under prolonged summer drought, and the mountainous interior by beech, fir and spruce [29]. Between 1990 and 2023, wildfires affected 115,499 ha, roughly 8 % of the national territory, with individual seasons ranging from fewer than 150 ha to over 21,000 ha in 2017 [32]. This combination of an optically obstructed landscape, extensive forest cover and episodic seasons of very large burned area defines the conditions under which the surveillance potential of the existing antenna tower network is assessed.

3 Materials and methods

The analysis evaluated the wildfire surveillance potential of the existing antenna tower network operated by One Crna Gora using viewshed and cumulative viewshed modelling. The overall workflow is presented in Figure 2.

3.1 Input data

Viewshed analysis was performed using Copernicus DEM datasets available at three spatial resolutions: 10, 30 and 90 m. It should be emphasised that, despite its name, the Copernicus DEM is a DSM, that is, it represents the reflective surface of the Earth including buildings, infrastructure and vegetation canopy, rather than a bare-earth terrain model. This distinction is methodologically relevant for the present study, because line-of-sight computation performed on a DSM implicitly accounts for the obstruction caused by forest canopy and built structures, which are the dominant above-ground barriers to smoke detection in Mediterranean and mountainous landscapes.

The Copernicus DEM is derived primarily from data acquired by the TanDEM-X mission between 2011 and 2015, with a specified absolute vertical accuracy better than 4 m (LE90) and absolute horizontal accuracy better than 6 m (CE90) [33]. Elevations are referenced to the EGM2008 geoid, and horizontal coordinates to WGS84 (G1150). The model is distributed in three instances: EEA-10 (10 m, covering the EEA39 countries), GLO-30 and GLO-90 (30 and 90 m, global coverage). Each instance is available in DGED and DTED product variants, which differ in the encoding of elevation values: DGED products store elevation as 32-bit floating-point values, whereas DTED products apply integer quantisation at 1 m vertical resolution. Since vertical quantisation directly influences line-of-sight computation over gently inclined terrain, the DGED variants were used for all viewshed computations, while the corresponding INSPIRE and DTED products were used only for verification of dataset consistency.

Antenna tower locations were obtained from the register published by One Crna Gora d.o.o. under the regulatory obligation of shared use of electronic communications infrastructure [34]. The register documents 111 locations, reporting for each the site name, municipality, coordinates of the tower base (WGS84), ground elevation, structural type and tower height. It was selected as the input dataset because it is the only publicly available national source documenting both the position and the structural characteristics of individual towers.

Four preprocessing steps were applied. First, coordinates were tested against the national bounding box (41.80–43.60° N, 18.40–20.40° E): two records with interchanged latitude and longitude were corrected and retained, while one record falling outside the national territory was excluded. Second, six records were excluded because the reported structure is not a free-standing tower capable of hosting a monitoring camera, namely antenna holders on building facades, roof-mounted frames, a street-lighting pole, and antennas installed in a road tunnel. Third, coordinates were reprojected to ETRS89/UTM zone 34 N (EPSG: 25834) and three exact positional duplicates were removed. Fourth, towers within 1 km of one another were treated as a single cluster, since their viewsheds are largely redundant at the analysed scales; within each cluster the most representative tower was retained, determined by the greatest structural height and, in the case of equal height, the greater ground elevation. This removed two further locations in central Podgorica. The final dataset comprises 99 antenna towers with structural heights of 9–48 m (median 30 m) and base elevations of 1–1,874 m a.s.l. (median 622 m); their spatial distribution is shown in Figure 1. The complete selection log for all 111 register entries is provided as supplementary material.

3.2 GIS processing and data preparation

All spatial analyses were performed within a fully automated workflow based on Python and the Geospatial Data Abstraction Library (GDAL), integrating GIS methods and

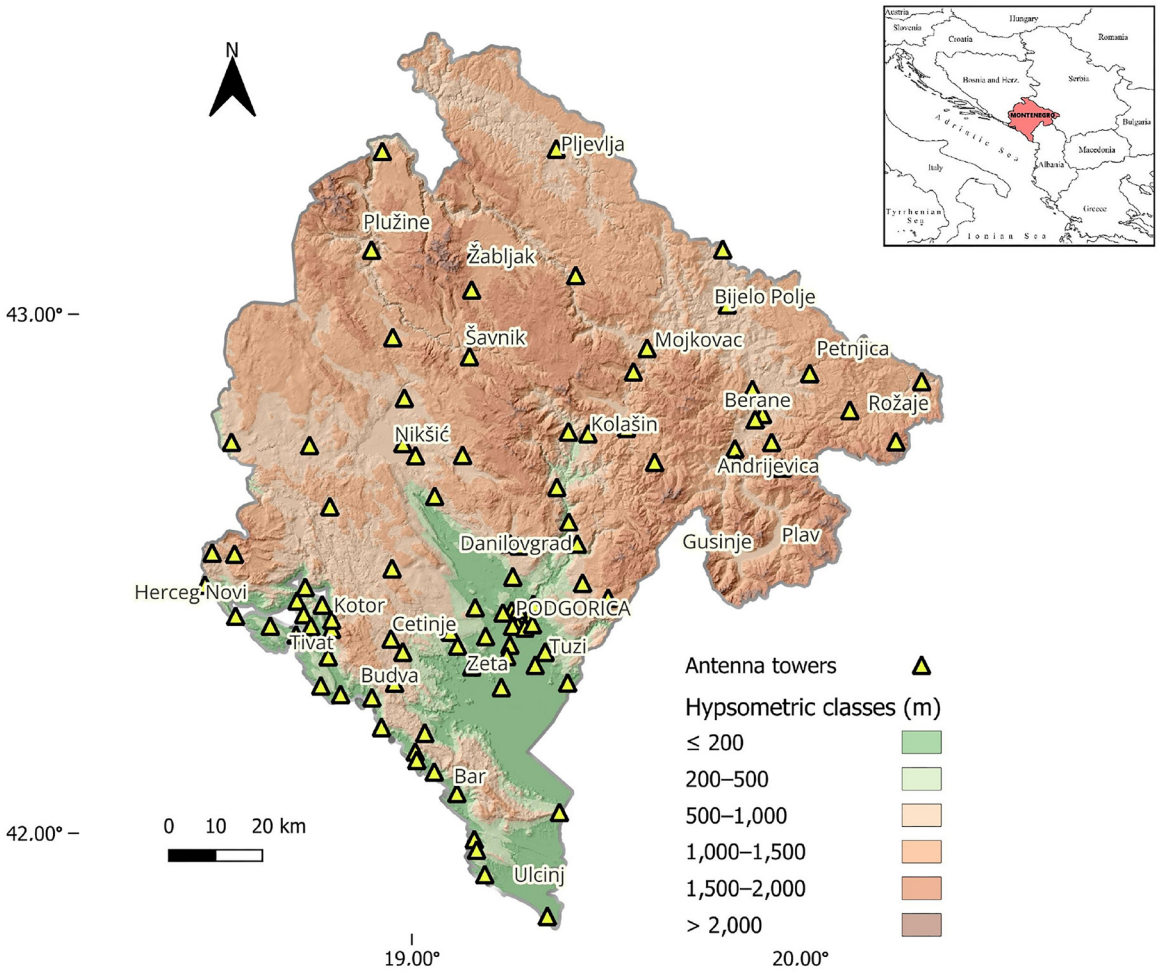


Figure 1: Location of Montenegro and spatial distribution of the 99 analysed antenna towers over hypsometric classes.

cumulative viewshed modelling. Inspection of the tower register and the preparation of the position map (Figure 1) were carried out in QGIS 3.40.9, whereas reprojection and clipping of the datasets, viewshed computation, raster algebra, coverage statistics and the analytical figures (Figures 3–6) were produced in Python 3.11 using the GDAL/OGR bindings, NumPy, pandas and Matplotlib. The national boundary of Montenegro was used to clip all raster datasets to the study area extent, and all DSM datasets and tower coordinates were reprojected into the European Terrestrial Reference System ETRS89/UTM zone 34 N (EPSG: 25834) in order to ensure spatial consistency during line-of-sight calculations.

The workflow comprised five sequential stages. First, the three Copernicus DSM datasets were reprojected and clipped to the national boundary, and cells outside the territory were assigned a NoData value. Second, tower coordinates were reprojected and converted to raster row and column indices, and towers falling outside the raster extent or on NoData cells were reported and excluded. Third, an individual viewshed was computed for every tower under each parameter combination. Fourth, the individual viewsheds of all towers were combined into a single cumulative visibility layer for that combination. Fifth, coverage statistics were derived and exported. The first stage was performed once per DSM resolution, whereas the third and fourth were repeated for each of the 108 parameter combinations, yielding a total of 10,692 individual viewshed computations.

Viewshed modelling was conducted with the GDAL gdal_viewshed utility [35], executed from Python through the GDAL bindings, which implements the sightline-free viewshed algorithm of Wang et al. [36]. Rather than tracing an explicit line of sight to every cell, the algorithm proceeds outward from the observer and derives, for each cell, the minimum elevation required for visibility from the already processed neighbouring cells located between that cell and the observer; a cell is classified as visible if its surface elevation, increased by the target height, meets or exceeds

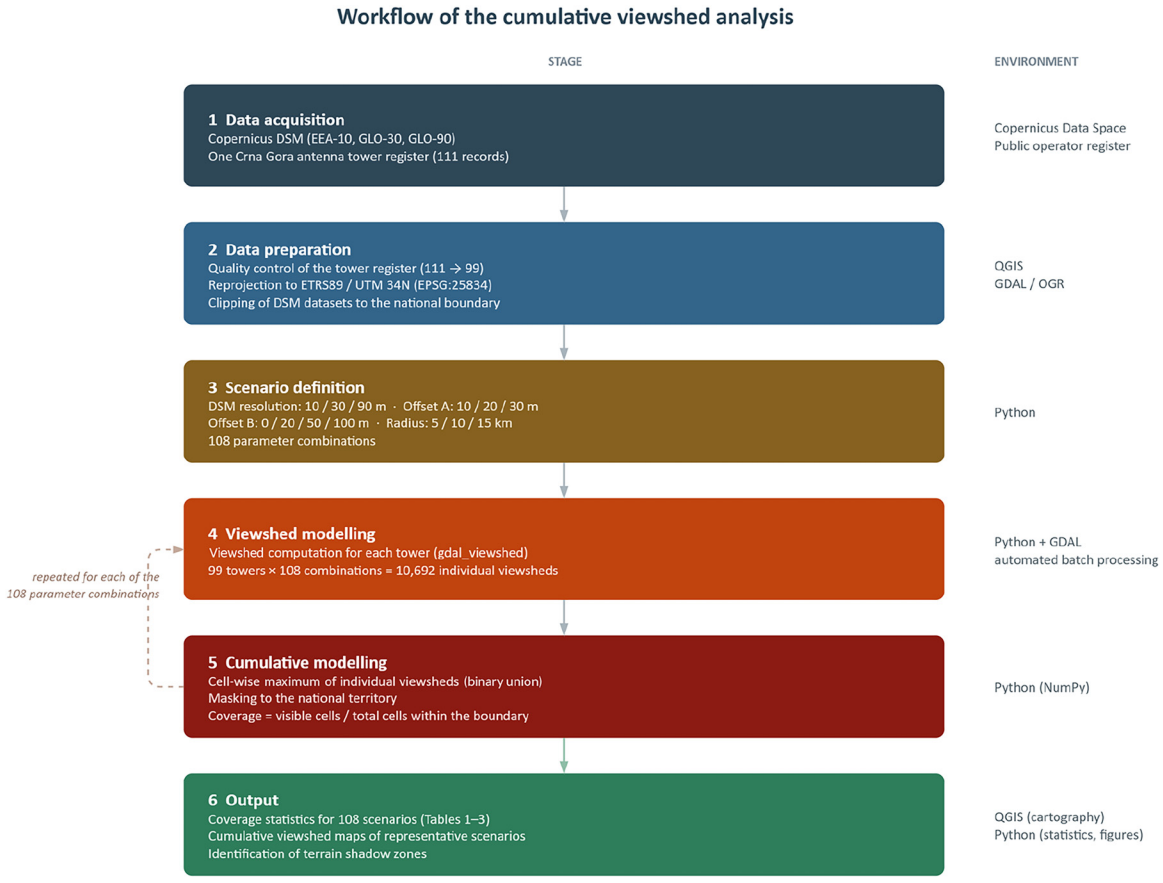


Figure 2: Workflow of the viewshed analysis methodology.

this threshold. Surface elevations are corrected for Earth curvature and atmospheric refraction through a curvature coefficient, and cells beyond the specified maximum distance are excluded from the computation.

Multiple viewshed scenarios were generated in order to simulate different wildfire detection conditions using the three DSM resolutions. Three scenario parameters were varied, corresponding directly to the gdal_viewshed parameters observer height, target height and maximum distance (command-line options -oz, -tz and -md). Throughout this paper they are denoted offset A, offset B and radius, respectively; this notation follows the observer-parameter convention of the ArcGIS Viewshed tool (OFFSETA, OFFSETB, RADIUS2) and is retained for comparability with previous fire-lookout visibility studies, but it refers to the scenario parameters only, all computations having been performed with GDAL. Observer height (offset A) defines the vertical distance of the camera above the surface represented by the DSM at the tower location, and was varied between 10, 20 and 30 m. This range corresponds to the structural heights documented in the tower register, in which most towers fall between 20 and 42 m, so that the three values represent a low, an intermediate, and an upper realistic mounting position on the existing infrastructure. The values were applied uniformly to all towers as scenario parameters rather than as the installed height of each individual structure, which allows the effect of mounting height to be isolated from variation in tower design.

Target height (offset B) defines the vertical distance above the surface at which a cell is considered detected, and was varied between 0, 20, 50 and 100 m. Offset B = 0 m corresponds to direct line-of-sight to the ground surface and therefore represents the most demanding case, equivalent to detecting flames or a nascent smoke column at the point of ignition. The remaining values approximate the vertical extent of smoke columns at successive stages of fire development: 20 m for an incipient surface fire, 50 m for an established fire with a developed convective column, and 100 m for a large, intense wildfire whose plume is visible well above the surrounding canopy and terrain. Since detection in operational camera-based systems relies on the smoke column rather than on the flame front, this parameter is the principal control on the realism of the assessment.

The visibility radius, that is, the maximum distance from the observer beyond which cells are excluded from

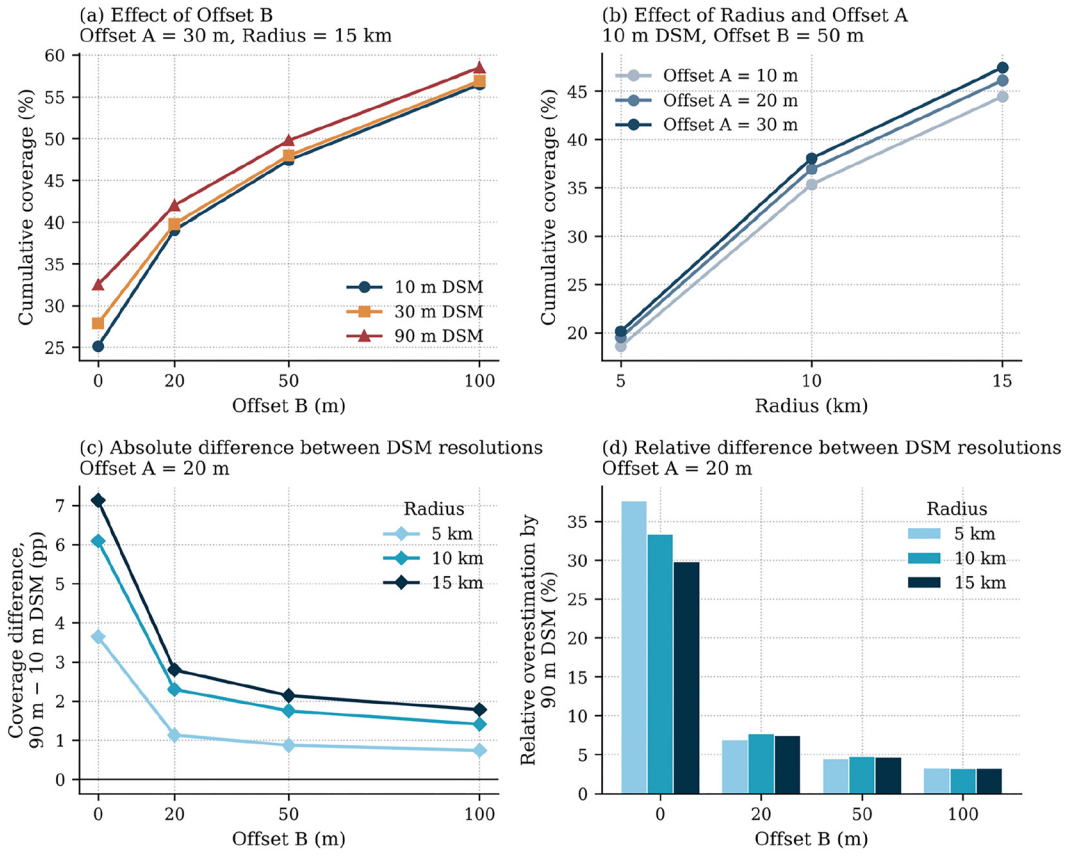


Figure 3: Effect of the analysed parameters on cumulative coverage: (a) effect of offset B; (b) effect of radius and offset A; (c) absolute and (d) relative difference between the 90 m and the 10 m DSM.

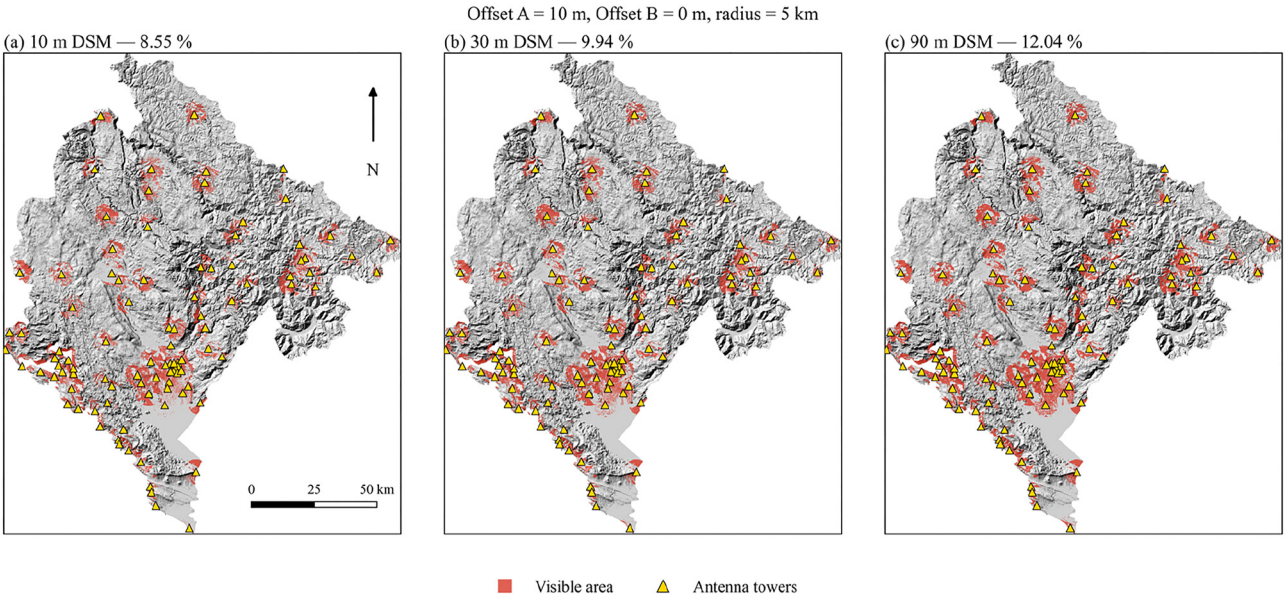


Figure 4: Cumulative viewshed under the most restrictive configuration (offset A = 10 m, offset B = 0 m, radius = 5 km): (a) 10 m, (b) 30 m and (c) 90 m DSM.

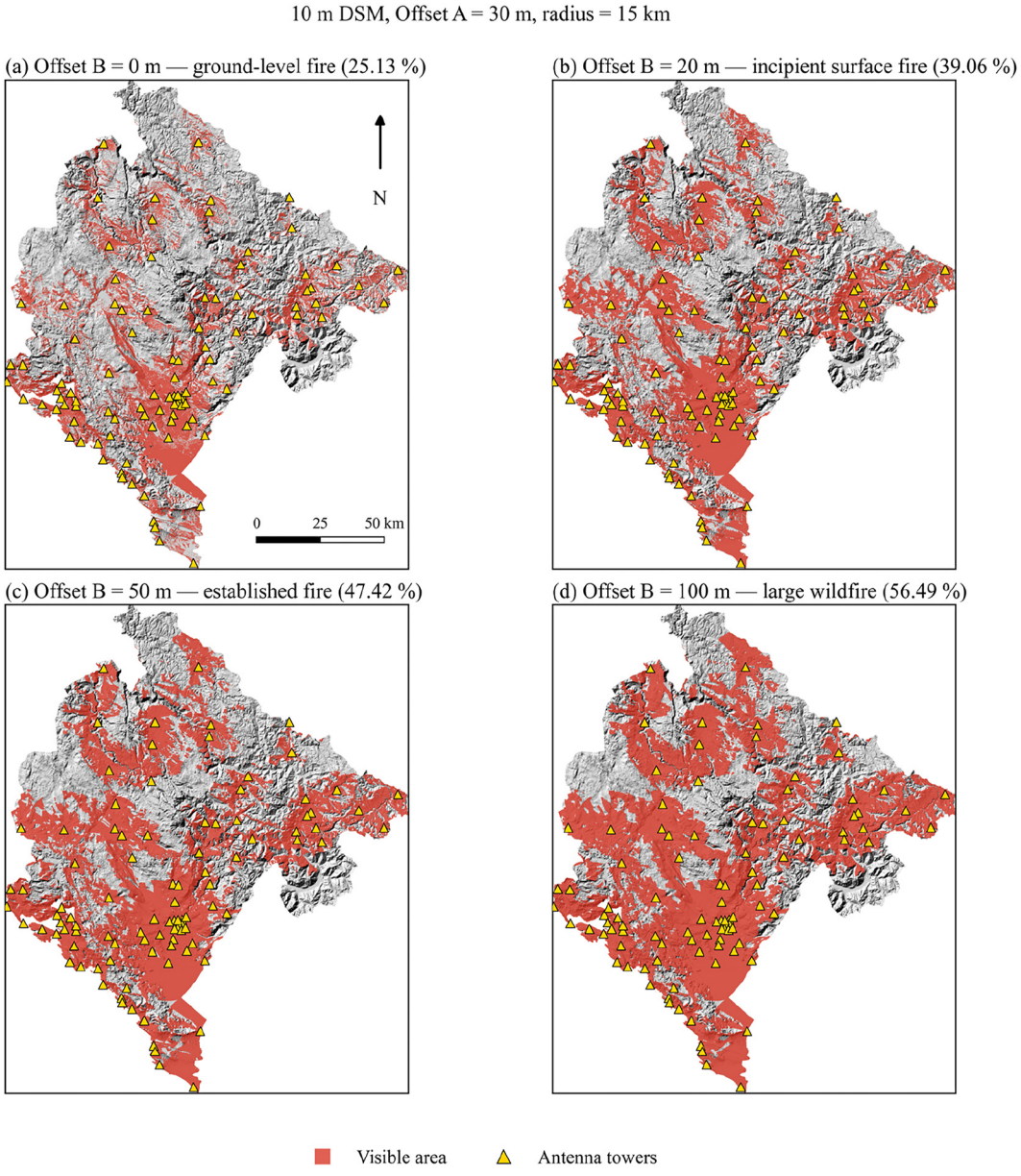


Figure 5: Cumulative viewshed on the 10 m DSM at successive wildfire development stages (offset A = 30 m, radius = 15 km): offset B = (a) 0 m, (b) 20 m, (c) 50 m and (d) 100 m.

the computation, was limited to 5, 10 and 15 km, reflecting the effective range over which commercial smoke-detection cameras can resolve a plume against the background under favourable atmospheric conditions. The lower bound represents conservative detection of small plumes, while 15 km approaches the upper operational limit reported for such systems; beyond this range, atmospheric attenuation rather than terrain becomes the limiting factor, so that purely geometric visibility modelling loses its interpretative value. All remaining parameters of gdal_viewshed were retained at their default values, including the curvature coefficient of 0.85714, which accounts jointly for Earth curvature and atmospheric refraction and is equivalent to a refraction coefficient of 1/7 (the curvature coefficient being defined as one minus the refraction coefficient), and the binary output convention in which visible cells are assigned a value of 255 and non-visible cells a value of 0. Observer and target heights are interpreted by the algorithm as offsets above the DSM surface rather than as absolute elevations.

For each parameter combination, the individual tower viewsheds were combined by cell-wise maximum, so that a

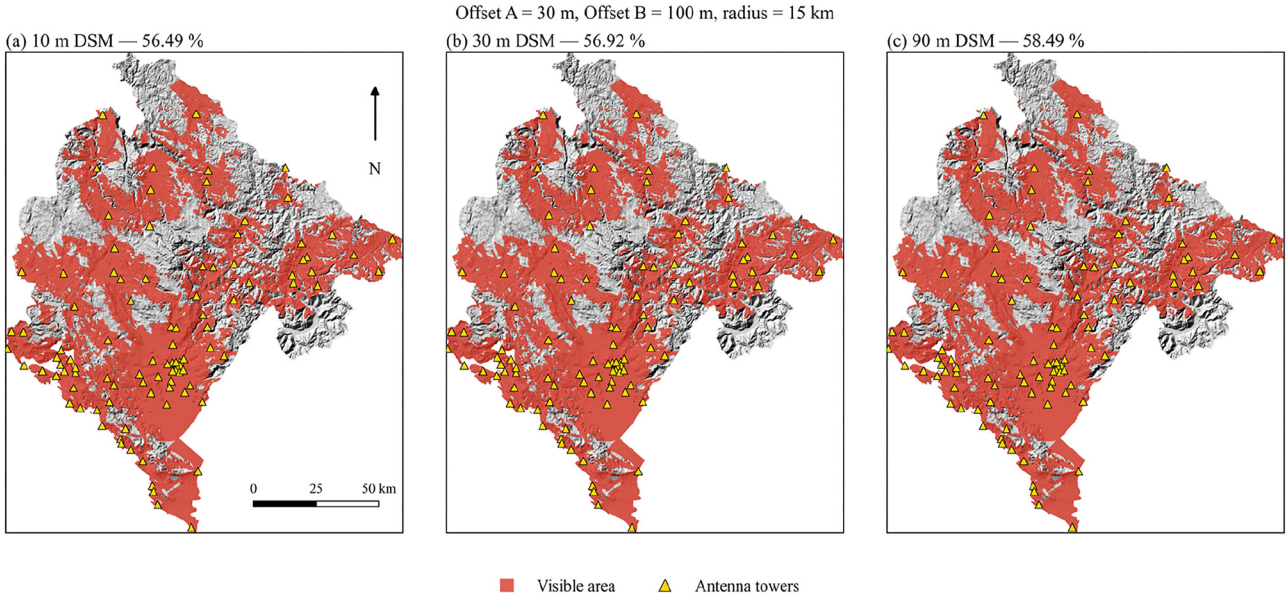


Figure 6: Cumulative viewshed under the most permissive configuration (offset A = 30 m, offset B = 100 m, radius = 15 km): (a) 10 m, (b) 30 m and (c) 90 m DSM.

cell was classified as visible in the cumulative layer if it was visible from at least one tower. The cumulative layer therefore represents the binary union of the individual viewsheds rather than a count of the towers from which each cell is visible. The resulting binary layer was then multiplied by the national territory mask in order to exclude cells outside the boundary from the statistics. Coverage was defined as the proportion of national territory visible from at least one tower under the given parameter combination, expressed as the number of visible cells divided by the total number of cells within the national boundary. This metric describes theoretical geometric visibility, that is, the existence of an unobstructed line of sight under the given observation geometry. It does not incorporate camera sensor characteristics, atmospheric visibility, smoke density or observer response, and therefore represents an upper bound on the detection capability of the network rather than an estimate of detection probability.

4 Results

The results of the conducted research were obtained through cumulative viewshed analysis performed for 99 antenna tower locations distributed across the territory of Montenegro. The cumulative viewshed analysis was performed using three Copernicus DSM spatial resolutions (10, 30, and 90 m), three observer heights (offset A = 10, 20, and 30 m), four target heights (offset B = 0, 20, 50, and 100 m), and three visibility radii (5, 10, and 15 km). A total of 108 cumulative viewshed scenarios were generated and analyzed. Table 1 presents the cumulative viewshed results obtained using the 10 m DSM, Table 2 shows the results generated using the 30 m DSM, while Table 3 presents the values calculated using the 90 m DSM.

Across all three DSM resolutions, cumulative visibility coverage increased monotonically with observer height, target height and visibility radius. The obtained values span a wide range, from 8.55 % under the most restrictive configuration (10 m DSM, offset A = 10 m, offset B = 0 m, radius = 5 km) to 58.49 % under the most permissive one (90 m DSM, offset A = 30 m, offset B = 100 m, radius = 15 km). Even under the latter scenario, more than 40 % of the national territory remained outside the line of sight of any analysed tower. The influence of the individual parameters and the behaviour of the differences between DSM resolutions are summarised in Figure 3.

Target height produced the strongest single effect on cumulative coverage (Figure 3a). On the 10 m DSM, at offset A = 30 m and a 15 km radius, coverage increased from 25.13 % at ground level to 39.06 % at a target height of 20 m, 47.42 % at 50 m and 56.49 % at 100 m. The transition from ground level to a 20 m smoke column alone accounted for 13.9 percentage points, exceeding the combined contribution of all

Table 1: Cumulative viewshed results for the 10 m DSM resolution.

Offset A [m]	Offset B [m]	Radius [km]	Coverage [%]
10	0	5	8.63
10	0	10	16.87
10	0	15	22.45
10	20	5	15.50
10	20	10	28.72
10	20	15	36.17
10	50	5	18.73
10	50	10	35.74
10	50	15	44.71
10	100	5	21.89
10	100	10	43.18
10	100	15	53.83
20	0	5	9.76
20	0	10	18.48
20	0	15	24.07
20	20	5	16.47
20	20	10	30.39
20	20	15	37.95
20	50	5	19.62
20	50	10	37.29
20	50	15	46.34
20	100	5	22.68
20	100	10	44.64
20	100	15	55.35
30	0	5	10.59
30	0	10	19.65
30	0	15	25.30
30	20	5	17.13
30	20	10	31.50
30	20	15	39.23
30	50	5	20.23
30	50	10	38.39
30	50	15	47.62
30	100	5	23.24
30	100	10	45.67
30	100	15	56.53

Table 2: Cumulative viewshed results for the 30 m DSM resolution.

Offset A [m]	Offset B [m]	Radius [km]	Coverage [%]
10	0	5	9.94
10	0	10	18.90
10	0	15	24.87
10	20	5	15.73
10	20	10	29.09
10	20	15	36.77
10	50	5	18.87
10	50	10	35.84
10	50	15	44.99
10	100	5	22.05
10	100	10	43.20
10	100	15	54.13
20	0	5	11.11
20	0	10	20.53
20	0	15	26.57
20	20	5	16.66
20	20	10	30.69
20	20	15	38.53
20	50	5	19.75
20	50	10	37.38
20	50	15	46.68
20	100	5	22.86
20	100	10	44.68
20	100	15	55.71
30	0	5	11.96
30	0	10	21.74
30	0	15	27.86
30	20	5	17.37
30	20	10	31.81
30	20	15	39.79
30	50	5	20.41
30	50	10	38.51
30	50	15	47.97
30	100	5	23.49
30	100	10	45.74
30	100	15	56.92

subsequent increments. The same pattern was obtained for the 30 and 90 m datasets, where coverage under the corresponding scenarios rose from 27.86 to 56.92 % and from 32.51 to 58.49 % respectively.

Visibility radius produced the second largest effect, but with clearly diminishing returns (Figure 3b). On the 10 m DSM, extending the radius from 5 to 10 km increased coverage by an average of 15.3 percentage points across all scenarios, whereas the further extension from 10 to 15 km added only 8.4 percentage points. The same non-linearity was present at all three resolutions and under all target-height configurations.

Observer height exerted the weakest influence among the three parameters. Increasing offset A from 10 to 30 m improved cumulative coverage in every scenario, but the average gain amounted to 2.4 percentage points on the 10 and 30 m datasets and 2.6 percentage points on the 90 m dataset, with a maximum improvement of 3.3 percentage points.

Differences between DSM resolutions were systematic in direction, with the coarser datasets consistently producing higher coverage, but their magnitude depended strongly on target height (Figure 3c and d). At ground level, the 90 m DSM exceeded the 10 m DSM by 3.5–7.4 percentage points, corresponding to a relative overestimation of 29–41 %. This margin decreased sharply as soon as the target was raised above the surface, falling to 1.1–3.1 percentage points at a target height of 20 m, 0.9–2.4 percentage points

Table 3: Cumulative viewshed results for the 90 m DSM resolution.

Offset A [m]	Offset B [m]	Radius [km]	Coverage [%]
10	0	5	12.04
10	0	10	22.60
10	0	15	29.24
10	20	5	16.55
10	20	10	30.93
10	20	15	38.98
10	50	5	19.52
10	50	10	37.25
10	50	15	46.65
10	100	5	22.64
10	100	10	44.39
10	100	15	55.49
20	0	5	13.31
20	0	10	24.33
20	0	15	31.03
20	20	5	17.50
20	20	10	32.36
20	20	15	40.55
20	50	5	20.40
20	50	10	38.68
20	50	15	48.26
20	100	5	23.43
20	100	10	45.73
20	100	15	57.03
30	0	5	14.21
30	0	10	25.69
30	0	15	32.51
30	20	5	18.25
30	20	10	33.63
30	20	15	41.99
30	50	5	21.17
30	50	10	40.03
30	50	15	49.78
30	100	5	24.08
30	100	10	47.03
30	100	15	58.49

at 50 m, and 0.7–2.0 percentage points at 100 m. Expressed in relative terms, the overestimation declined from approximately 41 % at ground level to approximately 3 % under developed-plume conditions. The absolute differences increased with visibility radius, averaging 1.6 percentage points at 5 km, 3.0 percentage points at 10 km and 3.6 percentage points at 15 km, while the relative differences remained largely stable across radius.

Cumulative viewshed maps for the representative scenarios are presented in Figures 4–6. Under the ground-level, short-range configuration (Figure 4), visibility is confined to isolated patches around individual towers, with continuous shadow zones extending over the karst interior and the deeply incised canyons; the progressive enlargement of these patches from the 10 m to the 90 m DSM illustrates spatially the overestimation quantified in Figure 3. Figure 5 shows the same network on the 10 m DSM as the target height increases from ground level to a 100 m smoke column, revealing how the isolated patches coalesce into contiguous coverage over the coastal belt and the central part of the country. Under the most permissive configuration (Figure 6), coverage becomes nearly indistinguishable between the three resolutions, confirming the convergence identified in the tabulated results, while shadow zones persist in the northern high-mountain region and in the enclosed karst depressions of the west.

5 Discussion

The comparative analysis of cumulative viewshed results obtained using the 10, 30, and 90 m Copernicus DSM datasets revealed consistent spatial visibility patterns across all analysed scenarios, although several important differences related to terrain representation and wildfire detection conditions were identified. In all three resolutions, cumulative visibility coverage increased systematically with higher observer heights, larger target heights, and greater visibility radii. These trends were observed throughout all 108 analysed scenarios and indicate the strong influence of observation geometry and wildfire development stage on visibility performance within mountainous terrain.

The most pronounced differences between DSM resolutions were observed in scenarios characterized by low target heights and shorter visibility radii. Under these conditions, visibility calculations are strongly controlled by local terrain morphology, including ridges, valleys, slopes, and micro-topographic barriers that directly obstruct line-of-sight calculations. Consequently, the 10 m DSM consistently produced lower cumulative visibility values compared to the coarser datasets. These results indicate that finer resolutions provide a more detailed and terrain-realistic representation of visibility obstruction, particularly in highly heterogeneous mountainous landscapes such as Montenegro, whereas coarser datasets smooth terrain irregularities and reduce the representation of smaller barriers, resulting in overestimated visibility coverage.

Similar observations have been reported in previous studies. Mistick et al. [21] documented the same direction of bias when comparing spatial resolutions in mountainous terrain, attributing it to the loss of micro-topographic barriers under grid coarsening, while Šiljeg et al. [16] reached a comparable conclusion for fire-prevention viewshed modelling in

the karst landscape of Zadar County, morphologically similar to Montenegro. The relative overestimation obtained here at ground level, reaching approximately 41 % for the 90 m DSM, lies at the upper end of the published range. This is consistent with the exceptional relief fragmentation of the Montenegrin karst, where dolines, uvalas and rocky crests operating at scales of tens of metres are precisely the features that a 90 m grid cannot resolve, and it suggests that resolution effects reported for less fragmented terrain should not be transferred to karst environments without verification.

However, as the target height increased from ground level to elevated wildfire smoke plume scenarios, the differences between resolutions became progressively smaller. Under the most extensive analysed scenario, cumulative visibility coverage values became relatively similar across all datasets, reaching 56.49 % for the 10 m DSM, 56.92 % for the 30 m DSM, and 58.49 % for the 90 m DSM, with a total difference of approximately 2 percentage points. This pattern suggests that as wildfire smoke plumes become higher and more spatially dominant, local terrain barriers exert a reduced influence on overall visibility calculations. Elevated smoke plumes extend above smaller topographic obstructions, allowing line-of-sight calculations to become less sensitive to spatial detail.

The dependence of the resolution effect on target height, quantified here across the full parameter space, has to our knowledge not previously been reported in a wildfire surveillance context, and represents the principal methodological contribution of this study. Its operational implication is direct. Where the objective is monitoring of developed smoke plumes, a freely available 90 m global dataset yields coverage estimates within approximately 3 % of those obtained from a 10 m national dataset, at a fraction of the storage and processing requirements, which is relevant for countries without access to high-resolution elevation data. Where the objective is early detection of ignitions at ground level, the choice of resolution determines whether coverage is estimated at 8.6 or 12.0 %, a difference that would materially alter conclusions regarding the adequacy of the existing network.

The analysis additionally demonstrated that visibility radius exerts a strong influence on cumulative visibility coverage regardless of resolution. Nevertheless, the increase was not linear, indicating that terrain obstruction continues to limit visibility even under extended operational ranges. This finding corresponds with previous wildfire surveillance studies [19, 21], which reported that mountainous terrain remains one of the dominant limiting factors for line-of-sight wildfire detection systems despite improvements in observer height and sensor range.

Another important observation relates to the influence of observer height. Higher observation points reduce the impact of near-field terrain obstruction and improve horizon visibility, which is especially important in mountainous regions with strong topographic fragmentation. The comparatively weak magnitude of this effect nevertheless deserves emphasis, since increasing tower height is often assumed to be an effective remedy for insufficient coverage. In terrain where obstructions routinely exceed several hundred metres in vertical extent, an additional 20 m of camera elevation cannot compensate, and adding observation points at complementary locations is therefore a more effective measure than raising existing ones.

Overall, the comparative analysis indicates that DSM spatial resolution significantly affects cumulative viewshed estimation under scenarios involving low wildfire intensity, small smoke plumes, and shorter visibility distances, while for scenarios representing developed wildfires with elevated smoke plumes and larger operational ranges the results become considerably more stable across resolutions. These findings confirm the importance of selecting resolution according to the operational objectives of wildfire monitoring systems and support the applicability of cumulative viewshed modelling for evaluating wildfire surveillance infrastructure in complex mountainous Mediterranean environments.

The obtained results indicate that the existing antenna tower network in Montenegro represents a suitable basis for the development of a national wildfire monitoring system. Due to their elevated and distributed locations, the analysed towers provide relatively high cumulative visibility coverage across different parts of the country, despite the complex mountainous terrain, with particularly high values achieved in scenarios simulating developed wildfires. It should be noted that, unlike the purpose-built fire lookout towers examined in most previous studies [11, 14, 15, 19, 22], the network analysed here was designed to maximise population coverage for telecommunications rather than terrain visibility. That such an infrastructure nonetheless covers more than half of the national territory under developed-plume conditions is a notable result, since it indicates that commercial telecommunication networks may serve as an essentially cost-free foundation for national wildfire surveillance. The same siting logic also explains the observed spatial pattern of coverage, which concentrates in the coastal belt and the central part of the country, and it means that towers cluster in zones of elevated human activity where ignition probability is highest.

Despite significant methodological advances, viewshed analysis for wildfire surveillance still faces several important limitations. The analysis is based on a binary visibility model, which oversimplifies real smoke detection conditions by neglecting factors such as atmospheric visibility, smoke

density, background contrast, and observer perception, as well as camera sensor characteristics including angular resolution, field of view and the sensitivity of automatic smoke-recognition algorithms. The coverage values reported here therefore represent an upper bound on the detection capability of the network rather than an estimate of detection probability. Unlike many previous analyses based on bare-earth models that do not account for vegetation, buildings, or other surface obstacles, the present study employed a surface model and therefore implicitly incorporates canopy and structural obstruction; however, the underlying TanDEM-X acquisitions date from 2011 to 2015, so subsequent forest growth, harvesting, burned-area regeneration and construction are not reflected in the model. Observer heights were applied uniformly to all towers as scenario parameters rather than as the installed height of each individual structure, and the technical feasibility of installing equipment on any specific tower, including structural load capacity, power supply, data connectivity and access, was not assessed. Furthermore, high-resolution viewshed analysis over large spatial extents remains computationally demanding, creating a persistent trade-off between spatial resolution, study extent, and computational feasibility.

Nevertheless, the obtained results also indicate that visibility coverage could be significantly improved through further optimization of the surveillance network and the identification of additional observation locations. Therefore, future research should focus on proposing new optimal monitoring locations and improving cumulative spatial coverage in terrain shadow zones. Despite these limitations, the findings of this study provide an important basis for the further development and planning of wildfire monitoring systems in Montenegro.

6 Conclusions

This study evaluated the wildfire surveillance potential of the existing antenna tower network in Montenegro using a multi-scenario cumulative viewshed approach. The analysis was performed using 99 antenna tower locations and three Copernicus DSM spatial resolutions (10, 30, and 90 m), combined with different observer heights, wildfire smoke plume scenarios, and visibility radii. A total of 108 cumulative viewshed scenarios were generated in order to assess how terrain representation and operational visibility parameters influence wildfire detection potential in complex mountainous environments.

The obtained results demonstrated that cumulative visibility coverage systematically increases with higher observer heights, larger target heights, and greater visibility radii, ranging from 8.55 to 58.49 % of the national territory depending on the parameter configuration. Target height proved to be the dominant control, with the transition from ground level to a 20 m smoke column adding almost 14 percentage points, while observer height contributed on average only 2.4 percentage points across its full range. The highest visibility values were achieved under scenarios simulating developed wildfires characterized by elevated smoke plumes, while the lowest coverage was obtained for scenarios representing small ground-level fires.

The analysis additionally confirmed that DSM spatial resolution significantly influences cumulative viewshed estimation, particularly under low target-height and short-distance scenarios where local terrain morphology strongly controls line-of-sight obstruction. Finer datasets provided more terrain-realistic and conservative visibility estimates, whereas coarser resolutions produced higher coverage due to terrain generalization effects. This effect proved to be conditional rather than uniform: the 90 m dataset overestimated coverage by up to 41 % relative to the 10 m dataset at ground level, but by only about 3 % once an elevated smoke plume was assumed. This conditionality represents the principal methodological contribution of the study and indicates that resolution requirements should follow from the intended detection objective rather than from a general preference for finer data.

The study also demonstrated that the existing antenna tower network in Montenegro represents a suitable infrastructure basis for wildfire surveillance applications. Due to their elevated and distributed locations, the analysed towers provide relatively high cumulative visibility coverage despite the highly fragmented mountainous terrain, and many of them are positioned near areas of increased human activity where wildfire occurrence is often strongly influenced by anthropogenic factors. This is particularly notable given that the network was originally designed for telecommunications rather than for terrain surveillance.

These results should nevertheless be understood as a preliminary spatial screening of geometric visibility potential rather than as an operational design. The computed coverage defines where detection is geometrically possible, not where it is probable, since vegetation dynamics, atmospheric conditions, smoke visibility, camera detection performance and the technical feasibility of equipment installation were not considered. Furthermore, the results also indicate that certain terrain shadow zones and insufficiently covered areas remain present across all configurations, suggesting that cumulative visibility coverage could be further improved through optimization of the surveillance

network and identification of additional observation locations.

Future research should therefore focus on spatial optimization methods, integration of wildfire susceptibility models, camera detection capabilities, vegetation influence, atmospheric conditions, and validation of the modelled visibility against recorded fire events. Overall, this study demonstrates the importance and applicability of multi-scenario cumulative viewshed modelling for wildfire monitoring system assessment. Subject to local calibration and independent validation, the presented approach may support future wildfire monitoring planning and spatial optimization not only in Montenegro, but also in other Mediterranean and mountainous regions exposed to increasing wildfire risk.

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